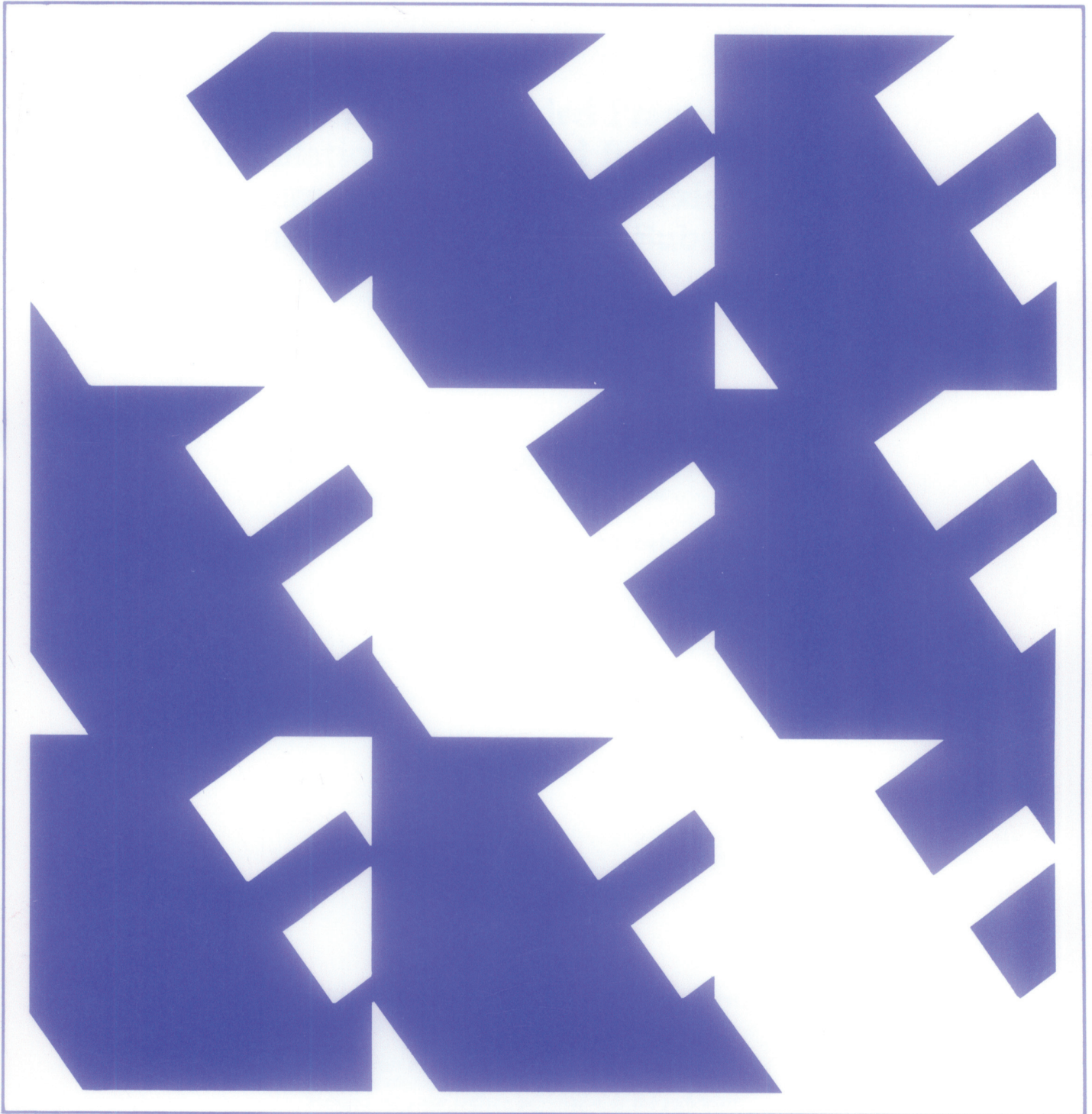


IEEE Recommended Practice for Installation Design and Installation of Large Lead Storage Batteries for Generating Stations and Substations



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of the
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Foreword

(This Foreword is not a part of IEEE Std 484-1981, IEEE Recommended Practice for Installation Design and Installation of Large Lead Storage Batteries for Generating Stations and Substations.)

Large stationary lead storage batteries play an ever increasing role in substation and generating station control and power systems and in providing the back-up energy for emergencies. This Recommended Practice fulfills the need within the industry to provide common or standard practices for the design of battery installations and the battery installation procedures. The methods described are applicable to all installations and battery sizes.

This Recommended Practice may be used separately, and, when combined with ANSI/IEEE Std 450-1980, IEEE Recommended Practice for Maintenance, Testing and Replacement of Large Lead Storage Batteries for Generating Stations and Substations, ANSI/IEEE Std 485-1978, IEEE Recommended Practice for Sizing Large Lead Storage Batteries for Generating Stations and Substations, will provide the user with a general guide to sizing, designing, placing in service, maintaining and testing a large storage battery installation. A companion standard, ANSI/IEEE Std 535-1979, provides a Standard for Qualification of Class 1E Lead Storage Batteries for Nuclear Power Generating Stations.

This Recommended Practice correlates and summarizes industry practices; it is not intended to be an exhaustive compilation or a rigid procedure manual.

The IEEE will maintain this Recommended Practice current with the state of the technology. Comments on this Recommended Practice and suggestions for additional material that should be included are invited. These should be addressed to:

Secretary

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IEEE Recommended Practice for Installation Design and Installation of Large Lead Storage Batteries for Generating Stations and Substations

1. Scope

This Recommended Practice provides recommended design practices and procedures for storage, location, mounting, ventilation, instrumentation, preassembly, assembly, and charging of large lead storage batteries. Required safety practices are also included. This Recommended Practice applies particularly to installations at generating stations, including nuclear, and large substations.

The portions of this Recommended Practice that will specifically relate to personnel and nuclear plant safety are mandatory and are designated by the word *shall*; all other portions are recommended practices and are designated by the word *should*.

Battery sizing, maintenance, capacity testing, charging equipment, dry charged cells, and consideration of other types of batteries are beyond the scope of this Recommended Practice.

2. Definitions

For definitions, refer to ANSI/IEEE Std 100-1977 [1]¹ and other documents referenced in Section 3.

3. References

[1] ANSI/IEEE Std 100-1977, Dictionary of Electrical and Electronics Terms

[2] ANSI/IEEE Std 344-1975, Recommended Practices for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations

¹Numbers in brackets correspond to those of the references listed in Section 3 of this Recommended Practice.

[3] ANSI/IEEE Std 485-1978, Recommended Practice for Sizing Large Lead Storage Batteries for Generating Stations and Substations

[4] ANSI/IEEE Std 494-1974, Standard Method for Identification of Documents Related to Class 1E Equipment and Systems for Nuclear Power Generating Stations

[5] ANSI/IEEE Std 535-1979, Standard for Qualification of Class 1E Lead Storage Batteries for Nuclear Power Generating Stations

[6] IEEE Std 308-1980, Standard Criteria for Class 1E Power Systems for Nuclear Power Generating Stations

[7] IEEE Std 336-1980, Installation, Inspection, and Testing Requirements for Class 1E Instrumentation and Electric Equipment at Nuclear Power Generating Stations

[8] IEEE Std 384-1981, Standard Criteria for Independence of Class 1E Equipment and Circuits

[9] IEEE Std 450-1980, Recommended Practice for Maintenance, Testing, and Replacement of Large Lead Storage Batteries for Generating Stations and Substations

[10] *Illuminating Engineering Society Handbook*.²

4. Safety

The safety precautions listed herein shall be followed during station battery installation. Work performed on batteries shall be done only with proper and safe tools, and the protective equipment listed below. Station bat-

²This document is available from the Illuminating Engineering Society, 345 E. 47th St., New York, NY 10017.

tery installation shall be performed or supervised by personnel knowledgeable of batteries and the safety precautions required.

4.1 Protective Equipment. The following equipment for safe handling of the battery and protection of personnel shall be available.

- (1) Goggles
- (2) Acid-resistant gloves
- (3) Protective aprons and overshoes
- (4) Portable or stationary water facilities for rinsing eyes and skin in case of acid spillage
- (5) Bicarbonate of soda or other suitable neutralizing agent recommended by the manufacturer for acid spillage
- (6) Lifting devices of adequate capacity, when required
- (7) Connector bolt tools with insulated handles

4.2 Procedures. The following safety procedures shall be followed prior to, and during installation:

- (1) Connect metal racks to station ground system
- (2) Inspect all lifting equipment for functional adequacy
- (3) Restrict all unauthorized personnel from the battery area
- (4) Prohibit smoking, arcing, and open flame in the immediate vicinity of the battery
- (5) Keep the top of the battery clear of all tools and other foreign objects
- (6) Ensure that illumination requirements are met
- (7) Ensure unobstructed egress from the battery area
- (8) Ensure battery area is ventilated during charging

5. Installation Design Criteria

Considerations that should be included in the design of the battery installation depend upon the requirements or function of the system of which the battery is a part.

5.1 General (All Installations). The following are the general installation design criteria for all large generating station and substation batteries, including Class 1E batteries for nuclear power generating stations.

5.1.1 Location

- (1) Space and floor supports allocated for the

battery and associated equipment should allow for present and future needs

- (2) The location should be as free from vibration as practical

- (3) The area selected should be clean, dry, well ventilated (see 5.1.4), and provide aisle space for inspection, maintenance, testing, and cell replacement. Space should also be provided above the cells to allow for operation of lifting equipment, addition of water, and taking measurements, (for example, temperature, specific gravity, etc).

- (4) The optimum cell electrolyte temperature, 77 °F (25 °C), is the basis for rated performance. A location where this temperature can be maintained will contribute to optimum battery life, performance, and cost of operation. Extreme ambient temperatures should be avoided because low temperatures decrease battery capacity while prolonged high temperatures shorten battery life

NOTE: Installation in a location with an ambient below the optimum temperature will affect sizing. Refer to IEEE Std 485-1978 [3].

- (5) The location or arrangement should result in no greater than a 5 °F (3 °C) temperature differential between cells at a given time. Localized heat sources such as direct sunlight, radiators, steam pipes, and space heaters should be avoided

- (6) Portable or stationary water facilities should be provided for rinsing spilled electrolyte. Provisions for neutralizing, containing, or safely dispersing spillage should be included

- (7) The charger and main power distribution center should be as close as practical to the battery, consistent with 5.1.1 (9)

- (8) Illumination in the battery area should equal or exceed the interior lighting recommendations in the *Illuminating Engineering Society Handbook* [10], Fig 9-80 for central station battery rooms

- (9) Nearby equipment with arcing contacts shall be located in such a manner as to avoid those areas where hydrogen pockets could form.

5.1.2 Mounting

- (1) The most common practice is to mount cells on a steel rack with acid-resistant insulation between the cells and the steel of the rack. Metal racks, if used, shall be connected solidly to the station grounding system. The cells may

also be mounted on insulated supports secured to a floor or base; the insulation used should be rated for full battery voltage. For seismic requirements refer to 5.2.2 (4) and (5)

(2) Not more than two tiers or two steps should normally be considered for large batteries; this choice of rack results in a minimum temperature differential between cells and will facilitate maintenance. A three tier rack is acceptable provided the requirements of maintenance and 5.1.1 (5) are met

(3) Cells in clear jars are usually mounted so that one edge of each plate is plainly visible for inspection

(4) Electrical connections to the battery and between cells on separate levels or racks should be made so as to reduce mechanical strain on battery posts. Refer to 5.2.2 (3)

5.1.3 Seismic. Although seismic consideration is a regulatory design requirement for nuclear power generating stations, fossil fuel generating stations and substations may appropriately follow the provisions of 5.2 when the installation is to be in a location subject to a high probability of seismic disturbance.

5.1.4 Ventilation. The battery area shall be ventilated, either by a natural or induced ventilation system, to prevent accumulation of hydrogen and to maintain design temperature. The ventilation system shall limit hydrogen accumulation to less than 2% of the total volume of the battery area. Maximum hydrogen evolution rate is 0.000269 cubic feet per minute per charging ampere per cell at 77 °F (25 °C), one atmosphere. The worst expected condition exists when forcing maximum current into a fully charged battery.

A battery area which meets the above ventilation requirements should be considered non-hazardous, thus special electrical equipment enclosures to prevent fire or explosions should not be necessary.

5.1.5 Instrumentation and Alarms. The following general recommendations for instrumentation and alarms apply to the battery installation only. Requirements for the charger, dc system design, etc, are beyond the scope of this Recommended Practice.

Each battery installation should include the following instrumentation and alarms:

- (1) Voltmeter
- (2) High and low battery voltage alarm
- (3) Ground detector (for ungrounded systems)

(4) Instrumentation to measure current through the battery. Refer to 4.5 of IEEE Std 450-1980 [9]

NOTE: The preceding recommendations for instrumentation and alarms could be satisfied by equipment in the dc system.

5.2 Nuclear Power Generation Station Class 1E Batteries. In addition to the general requirements presented in 5.1, a Class 1E battery, located within a nuclear power generating station, is subject to a number of additional design criteria. See IEEE Std 308-1980 [6], and ANSI/IEEE Std 535-1979 [5]. When the battery performs a Class 1E function, a quality assurance program shall be adopted to control and document all activities related to such functions.

5.2.1 Location

(1) The battery shall be protected against natural phenomena such as earthquakes, winds, and flooding, as well as induced phenomena such as fire, explosion, missiles, pipe whips, discharging fluids, CO₂ discharge, and other environmental hazards

(2) To minimize the effect of seismic forces, the battery should be located at as low an elevation as practical

(3) Where batteries are required in redundant systems, the batteries shall be separated as specified by IEEE Std 384-1981 [8]

5.2.2 Mounting

(1) All cells shall be restrained. Side and end rails with spacers between cells is one method that can be used to prevent loss of function due to a seismic event

(2) Cells shall be mounted with manufacturer's recommended separation distance and spacers. Any spacers used shall be moisture- and acid-resistant to avoid cell damage caused by spacer deformation

(3) Where more than one rack section is used, the rack sections shall be rigidly joined, or the adjacent end cells in each rack shall be connected with flexible connectors as provided by or recommended by the manufacturer. Connections between cells at different levels of the same rack shall also be flexible

(4) Racks shall be firmly connected to the building structure as specified by using approved fastening techniques such as embedded anchor bolts or racks welded to structural steel faceplates (sized to accommodate a range of battery rack sizes)

CAUTION: Anchoring a rack to both the floor and wall may cause stress due to conflicting modes of vibration.

(5) The racks, anchors, and installation thereof shall be able to withstand the force calculated for a safe shutdown earthquake to allow continuous battery service during and following that event, in accordance with ANSI/IEEE Std 344-1975 [2]

6. Installation Procedures

See Section 4 for safety precautions to be followed. In the case of Class 1E installation, refer to IEEE Std 336-1980 [7].

6.1 Receiving and Storage

6.1.1 Receiving Inspection. Upon receipt, and at the time of actual unloading, each package should be visually inspected for apparent damage and electrolyte leakage. If either is evident, a more detailed inspection of the entire shipment should be conducted and cell repair or replacement instituted as required. Record receipt date and inspection data.

6.1.2 Unpacking

(1) When lifting cells, a strap and strap spreader should be used, if applicable

(2) Always lift cells by the bottom, never by the cell posts

(3) Check electrolyte levels for evidence of leakage and that the plates are covered; any cell should be replaced if the electrolyte level is $\frac{1}{2}$ inch or more below the top of the plates

(4) All cells with visible defects such as cracked jars, loose terminal posts, or improperly aligned plates shall be repaired or replaced

6.1.3 Storage

(1) Cells should be stored indoors in a clean, level, dry, and cool location; extremely low ambient temperatures or localized sources of heat should be avoided

(2) Cells should not be stored for more than the time period recommended by the manufacturer, without applying a charge to the battery; in all cases, a period of three months storage is allowable between charges if the recommendations of (1) are followed

(3) For charging during storage or special conditions, the battery manufacturer should be consulted. Record dates and conditions for all charges during storage

6.2 Assembly

6.2.1 Rack Assembly. The assembly of the rack should be in accordance with the manufacturer's recommended procedure.

6.2.2 Cell Mounting and Connections. The following sequence should be used.

(1) Lift the individual cells onto the rack following the procedures outlined in 6.1.2 (1) and (2); mount the cells in accordance with the manufacturer's recommendations

(2) Where specified, install explosion-resistant vent plugs

(3) Check cell polarity for positive to negative connections throughout the battery

(4) Clean all terminal posts and connecting hardware areas showing evidence of corrosion, dirt, or acid, then coat all areas with a thin film of corrosion-inhibiting grease

(5) The intercell connector contact surfaces should be cleaned by rubbing gently with a brass suede brush or fine emery paper; care should be exercised in cleaning to prevent removal of the lead plating. Apply a thin film of corrosion-inhibiting grease to all contact surfaces

(6) Make intercell connections using manufacturer approved connectors (normally furnished with the battery)

(7) When more than one intercell connector per cell post is required, mount on opposite sides of post for maximum surface contact

(8) Tighten both ends of connection bolts to battery manufacturer's recommended torque value

(9) Clean all cell covers and containers; for dust and dirt use a water moistened clean wiper; for electrolyte spillage, use a bicarbonate of soda and water moistened wiper. All wipers used should be free of oil distillates

(10) For future identification, apply individual cell numbers in sequence beginning with number 1 at the positive end of the battery; also add any required operating identification

(11) Read the voltage of the battery to ensure that individual cells are connected correctly, that is, the total voltage should be approximately equal to the number of cells times the measured voltage of one cell. If the measurement is less, recheck the individual cell polarities

(12) Read and record intercell connection resistance to determine adequacy of initial installation and as a reference for future maintenance requirements (see Section 7, Records).

Review records of each intercell connection resistance measurement; remake and remeasure any connection that has a resistance measurement more than 10% over the average.

(13) When (1) through (12) have been satisfactorily completed, make final connections from the battery to the charger and dc system

6.2.3 Preoperational Care. After battery assembly is complete to the stage of intercell connections (see 6.2.2 (6)), the battery should be maintained in accordance with IEEE Std 450-1980 [9]. Protect the battery from dirt and debris.

6.3 Freshening Charge and Testing. Since a battery loses some of its charge during shipment and storage, a freshening charge should be applied after installation.

6.3.1 Freshening Charge Sequence

(1) Read and record the voltage, electrolyte temperature, and specific gravity of each cell prior to applying the charge; select the cell with the lowest specific gravity as a pilot cell

(2) Inspect all cells to ensure electrolyte level is above the top of the plates. Add only enough water to cover the plates.

(3) Follow manufacturer's recommendation for applying a freshening charge

(4) Read and record the specific gravity and temperature of electrolyte of the pilot cell at least once daily

(5) After pilot cell specific gravity readings become constant (with correction for temperature and level) return charger to float voltage

(6) At the end of 72 h, read and record all individual cell voltages, electrolyte temperatures, specific gravities, and levels. Correct the specific gravity for temperature and level and record the corrected values. (See Section 7, Records.) Refer to IEEE Std 450-1980 [9], Appendix A.

(7) Any cell that shows a specific gravity (corrected to 77 °F (25 °C) and to the high level line) less than 1.205 or more than 1.225, or a voltage lower than 2.13, requires corrective action in accordance with the manufacturer's instruction

NOTE: These criteria apply to a nominal 1.215 specific gravity battery. For other nominal specific gravity batteries refer to manufacturer.

(8) Add approved water to bring the electrolyte level of all cells up to the high level line.

6.3.2 Acceptance Test. When required, a constant current capacity discharge test shall be conducted in accordance with IEEE Std 450-1980 [9].

7. Records

Data obtained from receiving, storage, and assembly are pertinent to the maintenance and operational life of the battery. The data, that should be dated, recorded and maintained in a suitable permanent file for record purposes and future reference includes:

(1) Receiving inspection data and conditions of charge (see 6.1.3 (3))

(2) Initial resistance values of the intercell connections (see 6.2.2 (12))

(3) Individual cell specific gravity (corrected for temperature, voltage measurements, and electrolyte level) (see 6.3.1 (1), (4), (5), and (6))

(4) Acceptance test data (see 6.3.2)

(5) Quality assurance records for Class 1E batteries. Refer to IEEE Std 308-1980 [6], ANSI/IEEE Std 494-1974 [4], and ANSI/IEEE Std 535-1979 [5].

NOTE: The preceding records should be in accordance with cell identification (see 6.2.2 (10)).